Classification Model Evaluation for Breast Cancer Prediction

Task 4 – AI & Machine Learning Internship

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Task Title: Classification Model Evaluation
Dataset: Breast Cancer Dataset (Scikit-learn)

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1. Introduction

Classification is one of the most widely used machine learning techniques in real-world applications. Unlike regression tasks, classification focuses on predicting discrete labels or categories. In this task, the objective is to build a binary classification model and evaluate it using multiple performance metrics beyond accuracy.

The task emphasizes understanding classification metrics such as precision, recall, F1-score, and ROC-AUC, along with handling class imbalance and comparing different models. The goal is to select a reliable model based on comprehensive evaluation rather than relying on a single metric.

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2. Dataset Description

The dataset used for this task is the **Breast Cancer Dataset**, available in the Scikit-learn library. It contains features computed from digitized images of breast mass tissue samples.

Dataset Characteristics

- Total samples: **569**
- Number of features: **30**
- Target variable: Binary classification
    - **0 → Malignant**
    - **1 → Benign**

The dataset is clean, numerical, and does not contain missing values, making it suitable for classification modeling without additional preprocessing.

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3. Data Preparation

3.1 Train-Test Split

The dataset was split into:

- **80% training data**
- **20% testing data**

Stratified sampling was used to maintain class distribution in both sets.

3.2 Feature Scaling

StandardScaler was applied to normalize feature values. This ensures that all features contribute equally to model learning, especially for Logistic Regression.

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4. Baseline Model: Logistic Regression

A Logistic Regression model was trained as the baseline classifier.

4.1 Confusion Matrix

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[[41  1]
 [ 1 71]]
```

Interpretation

- **True Positives (71):** Correctly predicted benign cases
- **True Negatives (41):** Correctly predicted malignant cases
- **False Positives (1):** Incorrectly predicted benign
- **False Negatives (1):** Missed malignant case

The very low number of errors indicates strong model performance.

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5. Classification Metrics

Classification Report

- Precision ≈ **0.98–0.99**
- Recall ≈ **0.98–0.99**
- F1-score ≈ **0.98–0.99**
- Accuracy ≈ **98%**

Metric Interpretation

- **Precision:** High precision indicates very few false positive predictions
- **Recall:** High recall ensures most actual cases are correctly identified
- **F1-score:** Balanced measure combining precision and recall

In medical diagnosis, **recall is especially important**, as missing a disease case (false negative) can have serious consequences.

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6. ROC Curve and AUC Score

The ROC curve was plotted to evaluate model performance across different thresholds.

- **AUC Score ≈ 0.995**

Interpretation

An AUC score close to 1 indicates excellent model performance. The model is highly capable of distinguishing between malignant and benign cases.

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7. Handling Class Imbalance

To address potential class imbalance, Logistic Regression was retrained using:

class_weight = "balanced"

Balanced Model Results

- Accuracy ≈ **95–96%**
- Slight decrease in precision
- More balanced recall across classes

Interpretation

The balanced model prioritizes fairness between classes, slightly reducing accuracy but improving class-wise balance.

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8. Decision Tree Model Comparison

A Decision Tree Classifier was trained for comparison.

Results

- Accuracy ≈ **91%**
- Lower precision and recall compared to Logistic Regression

Observation

- Decision Tree shows weaker generalization
- More prone to overfitting compared to Logistic Regression

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9. Model Comparison Summary

Model	Accuracy
Logistic Regression	**0.982**
Logistic (Balanced)	0.956

Model	Accuracy
Decision Tree	0.912

Observation

- Logistic Regression clearly performs the best
- Balanced model trades some accuracy for fairness
- Decision Tree performs comparatively lower

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10. Final Model Selection

The **Logistic Regression model** was selected as the final model due to:

- Highest overall accuracy
- Strong precision and recall
- Excellent ROC-AUC score
- Stable and consistent performance

While the balanced model improves class fairness, the baseline Logistic Regression already provides high recall and precision, making it sufficient for this dataset.

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11. Conclusion

In this task, a binary classification system was developed and evaluated using multiple metrics. Logistic Regression demonstrated strong performance across all evaluation measures, including confusion matrix, classification report, and ROC-AUC.

This task highlights the importance of using multiple evaluation metrics rather than relying solely on accuracy. It also shows how model selection should consider both performance and reliability, especially in sensitive domains such as medical diagnosis.

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12. Future Scope

- Apply ensemble models like Random Forest
- Perform hyperparameter tuning for classification models
- Explore techniques like SMOTE for handling imbalance